NLP Assignment 2: RNN

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1 PPL 详细训练日志和测试结果

1.1 Settings

Parameter	Value
Epochs	50
Batch Size	20
BPTT	50
Embedding Dimension	32
Hidden Dimension	64
Learning Rate	0.001

Table 1: RNN Training Hyperparameters

1.2 训练日志

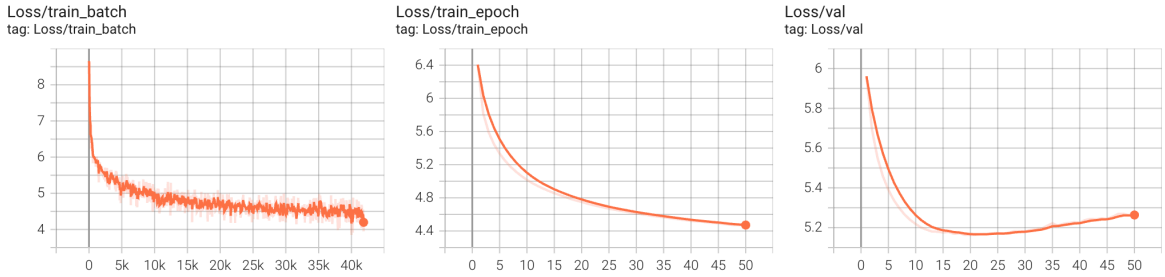


Figure 1: Training and validation loss over epochs

由 Figure 1可以发现，训练在 Epoch 超过 30 的时候就出现了过拟合的请款，最终我们选择了训练 Epoch 数为 30 的 checkpoint 作为后续测试的 RNN 模型。

1.3 `PTB.Test` 上 PPL 训练结果

我们使用 `PTB`[1] 作为我们的测试集，一共有 3761 条测试用例。如 Figure 2 所示。

[1] https://www.kaggle.com/datasets/smit1212/ptb-training

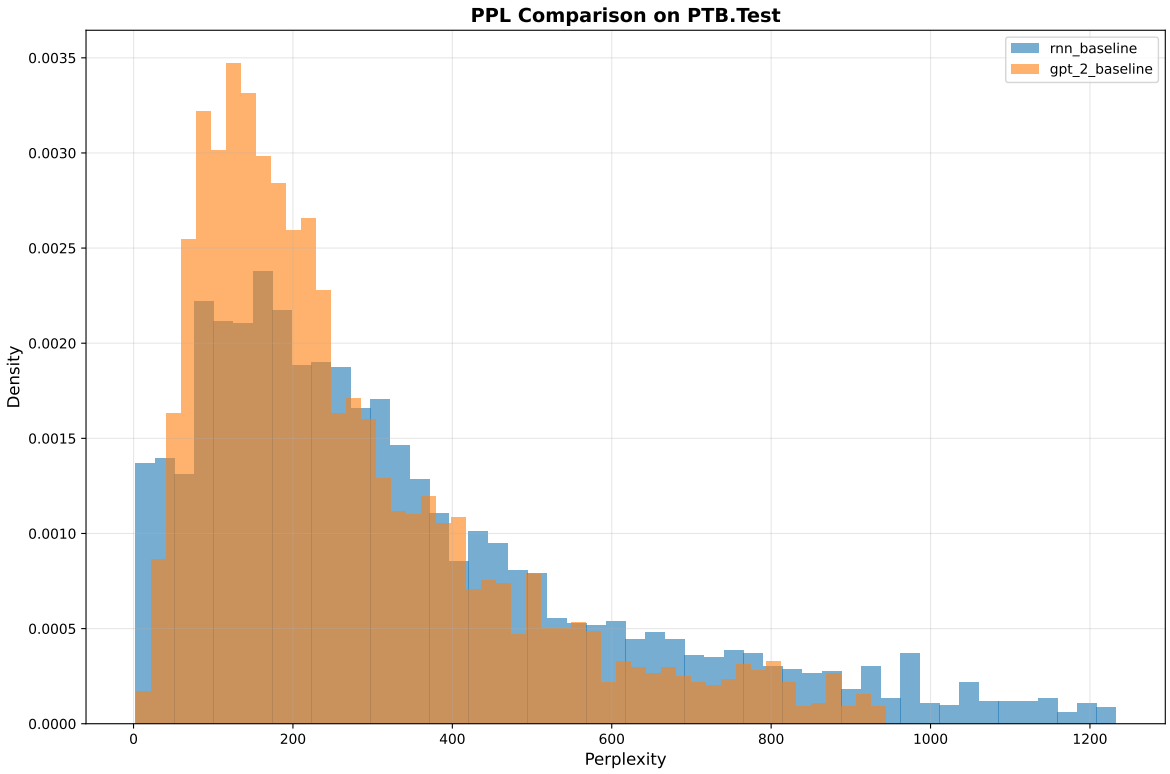


Figure 2: Distribution of perplexity scores on the PTB test set, comparing GPT-2 and RNN models.

$$L = -\sum_{t=1}^{5} \log\left(\mathrm{softmax}(y_t)_{x_{t+1}}\right)$$

1° 输出层.

$$\boxed{\delta_t = \frac{\partial L}{\partial y_t} = \mathrm{softmax}(y_t) - \mathrm{onehot}(x_{t+1})}$$ 误差信号

$$y_t = W_{out} h_t^{(2)} + b_{out}$$

$$\frac{\partial L}{\partial W_{out}} = \sum_{t=1}^{5} \frac{\partial L_t}{\partial W_o} \quad \left(L = \sum_{t=1}^{5} L_t\right)$$

$$= \sum_{t=1}^{5} \frac{\partial L_t}{\partial y_t} \frac{\partial y_t}{\partial W_o}$$

$$\frac{\partial L_t}{\partial y_t} = \frac{\partial L}{\partial y_t} = \delta \quad (\text{只有t一项包含 } y_t)$$

$$\frac{\partial y_t}{\partial W_o} = h_t^{(2)T}$$

$$\Rightarrow \frac{\partial L}{\partial W_{out}} = \sum_{t=1}^{5} \delta_t^{(out)} (h_t^{(2)})^T$$

$$\frac{\partial L}{\partial b_{out}} = \sum_{t=1}^{5} \frac{\partial L_t}{\partial b_o} = \sum_{t=1}^{5} \frac{\partial L_t}{\partial y_t} \boxed{\frac{\partial y_t}{\partial b_o}} \; I.$$

$$= \sum_{t=1}^{5} \delta_t^{(out)}$$

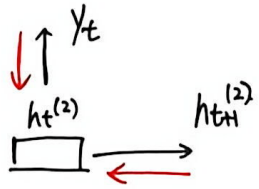
2° 第二层RNN

$$h_t^{(2)} = \tanh\left(W_{ih}^{(2)} h_t^{(1)} + W_{hh}^{(2)} h_{t-1}^{(2)} + b^{(2)}\right)$$

误差信号 $$\delta_t^{(2)} = \frac{\partial L}{\partial h_t^{(2)}} = \frac{\partial L}{\partial y_t} \cdot \frac{\partial y_t}{\partial h_t^{(2)}} + \frac{\partial L}{\partial h_{t+1}^{(2)}} \frac{\partial h_{t+1}^{(2)}}{\partial h_t^{(2)}}$$

$$= W_o^T \delta_t^{out} + (W_{hh}^{(2)})^T \delta_{t+1}^{(2)} \odot \left(1 - (h_{t+1}^{(2)})^2\right)$$

激活函数导数.



$$\frac{\partial L}{\partial W_{ih}^{(2)}} = \sum_{t=1}^{5} \frac{\partial L_t}{\partial W_{ih}^{(2)}}$$

$$= \sum_{t=1}^{5} \frac{\partial L_t}{\partial h_t^{(2)}} \underbrace{\frac{\partial h_t^{(2)}}{\partial z_t^{(2)}}}_{\odot (1-(h_t^{(2)})^2)} \cdot \frac{\partial z_t^{(2)}}{\partial W_{ih}^{(2)}}$$

$$= \sum_{t=1}^{5} \left(\delta_t^{(2)} \odot \left(1 - (h_t^{(2)})^2\right)\right) (h_t^{(1)})^T$$

$$\frac{\partial L}{\partial b_{out}} = \sum_{t=1}^{5} \left(\delta_t^{(2)} \odot \left(1 - (h_t^{(2)})^2\right)\right)$$

$$\frac{\partial L}{\partial W_{hh}^{(2)}} = \sum_{t=1}^{5} \left(\delta_t^{(2)} \odot \left(1 - (h_t^{(2)})^2\right)\right) (h_{t-1}^{(2)})^T$$

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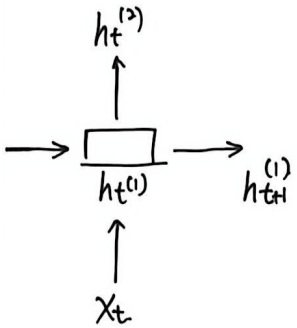
3° 第一层RNN.

误差信号 $\delta_t^{(1)} = \frac{\partial L}{\partial h_t^{(1)}} = \frac{\partial L}{\partial h_t^{(2)}} \cdot \frac{\partial h_t^{(2)}}{\partial h_t^{(1)}} + \frac{\partial L}{\partial h_{t+1}^{(1)}} \frac{\partial h_{t+1}^{(1)}}{\partial h_t^{(1)}}$

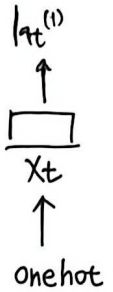
$$= (W_{zh}^{(2)})^T \delta_t^{(2)} \odot (1-(h_t^{(2)})^2) + (W_{hh}^{(1)}) \delta_{t+1}^{(1)} \odot (1-(h_{t+1}^{(1)})^2)$$



$$\begin{cases} \frac{\partial L}{\partial W_{zh}^{(1)}} = \sum_{t=1}^{5} \frac{\partial L_t}{\partial W_{zh}^{(1)}} \\ \quad = \sum_{t=1}^{5} \frac{\partial L_t}{\partial h_t^{(1)}} \cdot \frac{\partial h_t^{(1)}}{\partial z_t^{(1)}} \cdot \frac{\partial z_t^{(1)}}{\partial W_{zh}^{(1)}} \\ \quad = \sum_{t=1}^{5} \left(\delta_t^{(1)} \odot (1-(h_t^{(1)})^2) \right) x_t^T \\ \frac{\partial L}{\partial W_{hh}^{(1)}} = \sum_{t=1}^{5} \left(\delta_t^{(1)} \odot (1-(h_t^{(1)})^2) \right) (h_{t-1}^{(1)})^T \end{cases}$$

$$\frac{\partial L}{\partial b^{(1)}} = \sum_{t=1}^{5} (\delta_t^{(1)} \odot (1-(h_t^{(1)})^2)).$$

4° Embedding层.



$$\delta_t^{(e)} = \frac{\partial L}{\partial x_t} = \frac{\partial L}{\partial h_t^{(1)}} \frac{\partial h_t^{(1)}}{\partial x_t} = (W_{zh}^{(1)})^T \delta_t^{(1)} \odot (1-(h_t^{(1)})^2)$$

$$\frac{\partial L}{\partial E} = \sum_{i=1}^{5} \frac{\partial L_i}{\partial E} = \sum_{i=1}^{5} \frac{\partial L_i}{\partial x_i} \cdot \frac{\partial x_i}{\partial E}$$

$x = E^T o$ （x是某个行向量）

$$= \sum_{i=1}^{5} \delta_t^{(e)} \cdot \underline{o_t^T}$$

独热向量.

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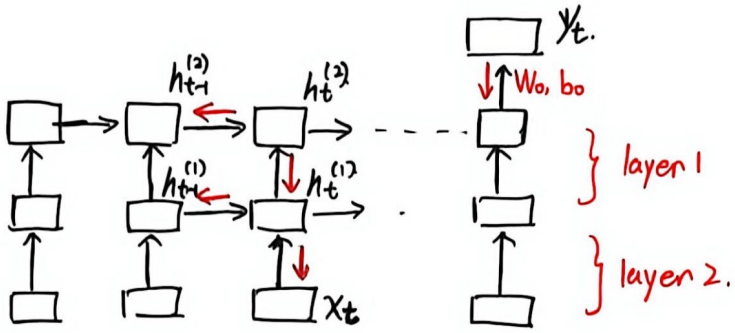

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